

Apps and Workflows to Support the Land Preservation Society of Norton (MA)

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Background

The Land Preservation Society of Norton (LPS) is a small, nonprofit land conservancy whose mission is to acquire and protect land in Norton, Massachusetts. Over the past fifty years, many land parcels have been donated or purchased for purposes of conservation. Recreational access has been established to a number of these locations through trails, on which hiking and biking are permitted year round. Over time the trails have been decently maintained, but poorly mapped and advertised. As a result, little awareness of their existence has spread through local communities.

The main departments within LPS are Land, Trails, and Outreach, and each is limited by number of personnel. The Land department focuses on acquiring property and maintaining land records of LPS-associated properties over time. While land records exist, they could stand to be organized and associated with existing map features. The focus of the Trails department is mostly focused on mapping and maintaining established trails. Many trails are decently mapped, but accuracy could be improved and updating line features may be simplified. The Outreach department focuses on connecting with local communities, especially engaging with students at local schools. Improving LPS visibility in Norton is of concern to the Outreach department. By using GIS to establish a way to engage users of LPS land and other local communities, department resources will be expanded, LPS land will be made more available, and the visibility of the LPS will be improved.

How GIS supports the mission of the LPS

During the last two years, I have been working to create LPS trail maps for public use. This work has helped to establish a crude baseline for GIS functionality within LPS and may be extended with geo-applications. The production of a suite of geo apps and workflows may improve the capabilities of each department within LPS by helping to document, organize, and connect information among them. The Trails department will benefit from apps that allow for marking up maps with notes, and for collecting trail data. For the Land department, geo apps that help to visualize LPS parcels, to scout potential future acquisitions, and to edit features will help to streamline the maintenance of land. Efforts of the Outreach department will benefit from apps that can be used both by LPS members and by the public to visualize trail maps and to report on visits to each trail site. There is significant overlap between the Land and Trails department in terms of the data they reference to perform tasks, and some apps will therefore serve to assist the goals of both departments. GIS is a powerful means of improving the mission of LPS and allows for the production of a suite of connected geo applications designed to advance the goals of each department. By using GIS, the LPS of Norton may update the quality of trail maps, improve the storage of geospatial land records, and more effectively communicate with local communities.

Introduction

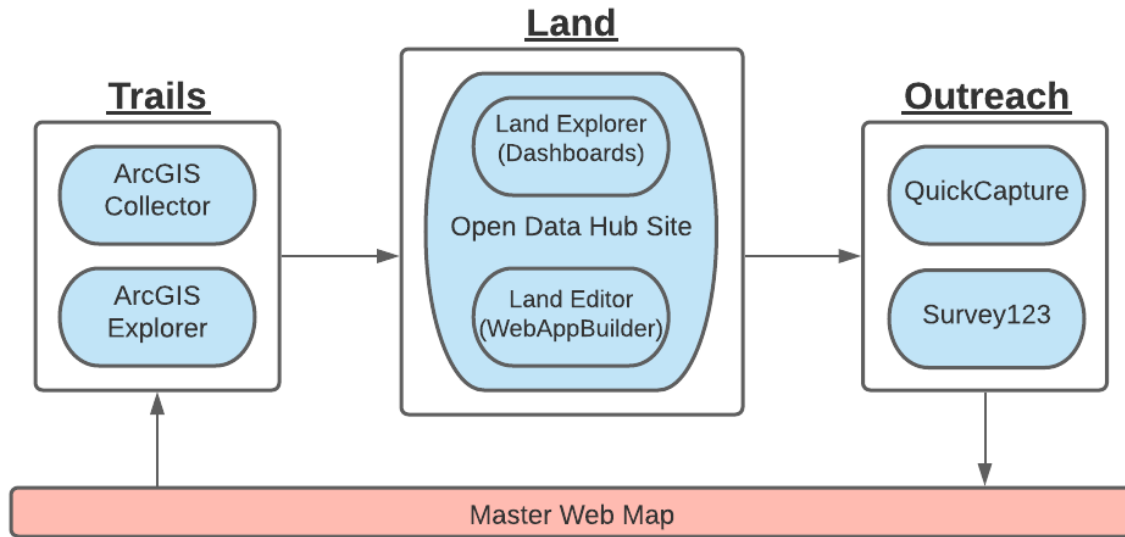


Diagram of apps, divided into bins based on the LPS departments they most highly benefit.

The apps are associated in a clockwise manner, starting and ending with the master web map that hydrates them. The master web map is to be opened in the Explorer and Collector apps under Trails, so edits and markups may be accessible by other applications. The data retrieved through these apps will then be accessible to the Land Explorer, and Land Editor apps under Land, available in an Open Data hub for easy access and shareability within the LPS. Land parcel and trail line data may also be configured in these apps. The QuickCapture and Survey123 apps under Outreach are kept updated with the data configured in the previous apps, and they subsequently update layers configured in the master web map.

The Trails department will benefit from apps which allow users to perform scouting tasks in the field by marking up trail maps with notes, and to collect line data to update trails. The Explorer for ArcGIS app will provide real-time scouting functionality, whereby users can markup maps with lines, polygons, and notes in the field, to be used for reference after the fact in Land Explorer and Land Editor apps. The Collector for ArcGIS app will allow Trails users to collect new line information on the trails. This lets users update trail locations with newly collected data.

The Land department will benefit from apps designed to edit and export queried parcel data, and to review statistics on trail usage. An Open Data Hub site will facilitate organizational access by containing these apps, and various other information products of concern to LPS. ArcGIS Dashboards was used to produce a Land Explorer that allows users to identify statistics on land attributes and usage to help target desirable land for acquisition. The Trails department may also use this app to review statistics on trail usage. Web App Builder was used to configure a Land Editor that allows users to scout potential land acquisitions, update tax parcel and point layers, and mark up the map with points lines and polygons for reference before going into the field to deploy the Explorer and Collector apps.

LPS Outreach will benefit from apps which help to connect LPS efforts and local communities. The QuickCapture app allows involved parties with an ArcGIS log in to collect point data on features that exist on LPS trails. These data may then be used to hydrate public-facing apps intended to help inform trail users on conditions and natural features. Such a configuration enables Trails members and student interns to collect important data on LPS trails. A Survey123 app presented at trail head signage will prompt all trail users to submit mobile feedback, further connecting the LPS and Norton communities.

Trails Department

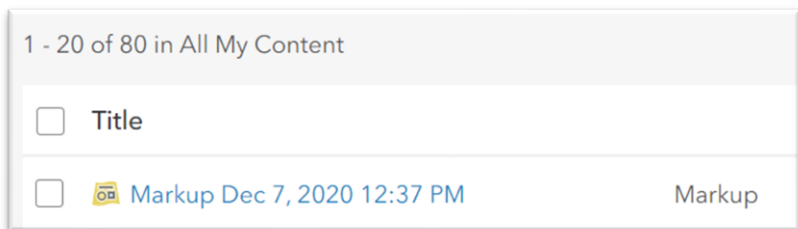
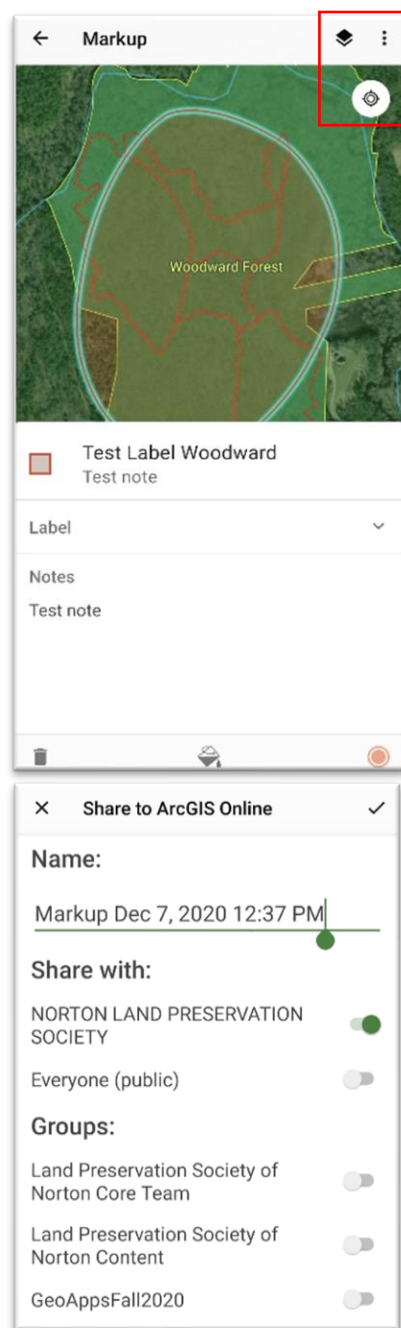
The Trails department of LPS is responsible for maintaining all aspects of the public hiking trails. The focus of this department is on managing trail signage and markers, updating trail maps, and maintaining the trails. These responsibilities rely on accurate and detailed location information on trail extents and points of interest and may greatly benefit from the integration of GIS applications. Apps which allow Trail department members to visualize and markup trail maps in real time and to collect new line and point data, will streamline the process of keeping trails, trail maps, and signs up to date. The Collector and Explorer for ArcGIS mobile application templates are best suited to power these workflows. These apps will be hydrated by a master web map produced in ArcGIS Pro and configured in ArcGIS Online.

Explorer for ArcGIS

The Explorer for ArcGIS mobile app template will support the Trails department by giving users a more casual data collection experience. None of the information collected using the Explorer app will be publishable. The app is used as a digital notepad that allows users to markup the map with observations in the field and add notes to the mark up features. This will assist users in marking locations to add trail signage, target areas in need of maintenance, and scout future trails. The collected information may be referred to later to improve trail quality and accuracy, and to produce concept maps of future trails.

Instructions

1. Download *Explorer for ArcGIS* and sign-in using LPS credentials.
2. Select *LPS Master Map* to load into the viewer.
3. While at an LPS location, press the *My Location* button in the upper right corner to pan to your location.
4. While on a trail your location will dynamically update – use *Layers* icon at the top of the screen to select and de-select viewable layers.
5. Use the pencil icon at the top of the screen to start the *Markup* function. Draw lines or shapes on the map to mark areas of interest, using your location as reference. You may add notes to the marks by tapping on them.
6. With the *Markup* function open, press the three dots on the upper right corner to share the Markup to ArcGIS Online, or share a screenshot how you wish. When sharing to ArcGIS Online, give your Markup a name, and press the check mark in the upper right corner.
7. View the Markup on ArcGIS Online for reference when working in a desktop environment.



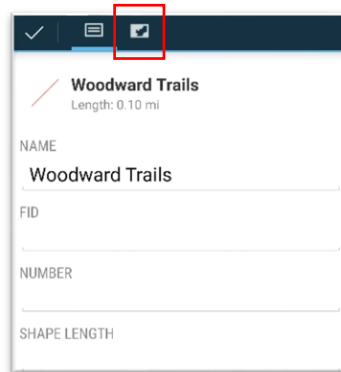
Collector for ArcGIS

The Collector for ArcGIS mobile app template will support the Trails department by allowing users to record trails in an easy way. It enables members of the Trails department to add new trails to the web map, and to update old trails to improve accuracy. The workflow will include using the Collector app to stream and collect live data, and then sending the collected data to the trails web layer to update the web map. The trail data may be configured and edited later either in the web map or using the Land Editor app. This tool helps to improve the Trails department's responsibility to keep trails up to date and may be relied upon to help update trail maps at trail head signage.

Instructions

1. Download *Collector Classic for ArcGIS* and sign-in using LPS credentials.
2. Select *LPS Master Map* to load into the viewer.
3. While at an LPS location, press the “My Location” button in the lower left corner to pan to your location, or select a bookmark to zoom to a site. If necessary zoom in until the trail is visible.
4. While on a trail your location will dynamically update – use the three dots in the bottom right corner of the screen to add or remove viewable layers, change the basemap, or measure distances for real-time reference.
5. To start recording a trail, select the “+” button at the bottom of the screen, and choose the trail you wish to continue. Only record trails or trail segments that do not appear on the map. This will prevent multiple trail lines from being drawn on top of each other.

6. After selecting the trail to record, leave the text fields as they are and press the Map icon at the top of the screen to obtain a map view. Note the GPS accuracy (in ft) in the bottom left corner of the map view.

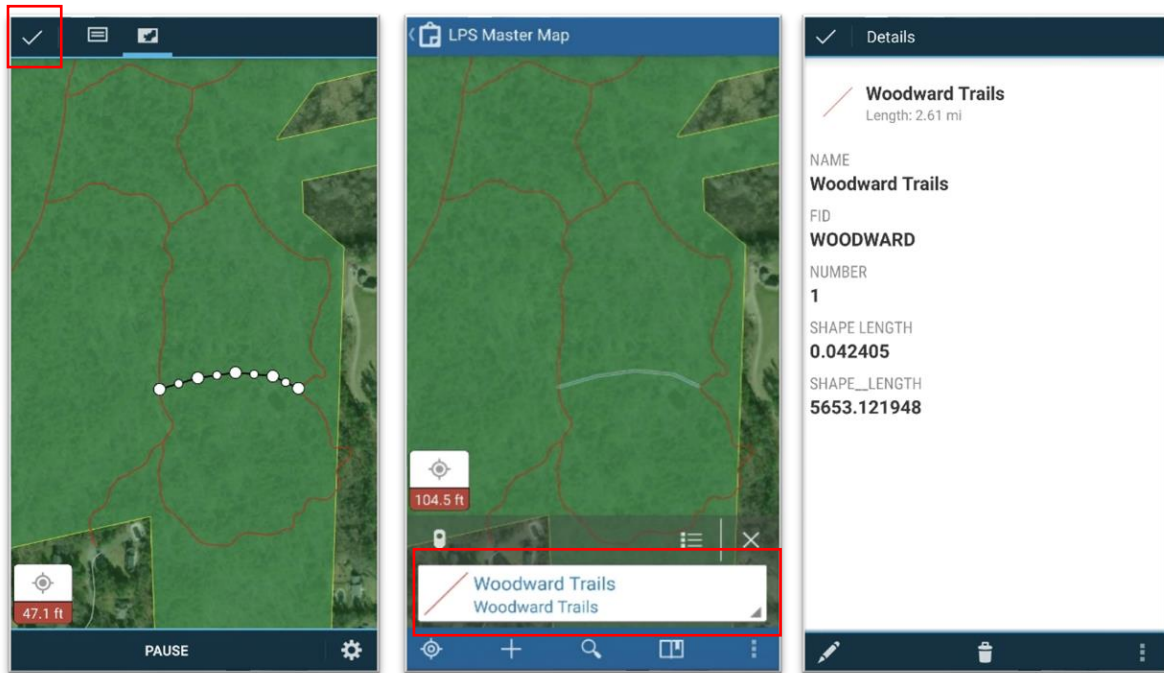


7. To change Collect settings, press the three dots in the bottom right corner of the screen. Ensure streaming interval is set to 5 seconds, and (if using only your phone's GPS sensor) that GPS averaging is OFF. Press OK. Press “STREAM” below the GPS accuracy readout, to start recording a line. While streaming is paused, you may also touch the map to add vertices.

Note: Only pause the stream when you are standing still. If you are collecting multiple trails that are not continuous, name and submit each non-continuous trail separately before recording the next trail. This will prevent recording over already-mapped line data in the selected trail layer.

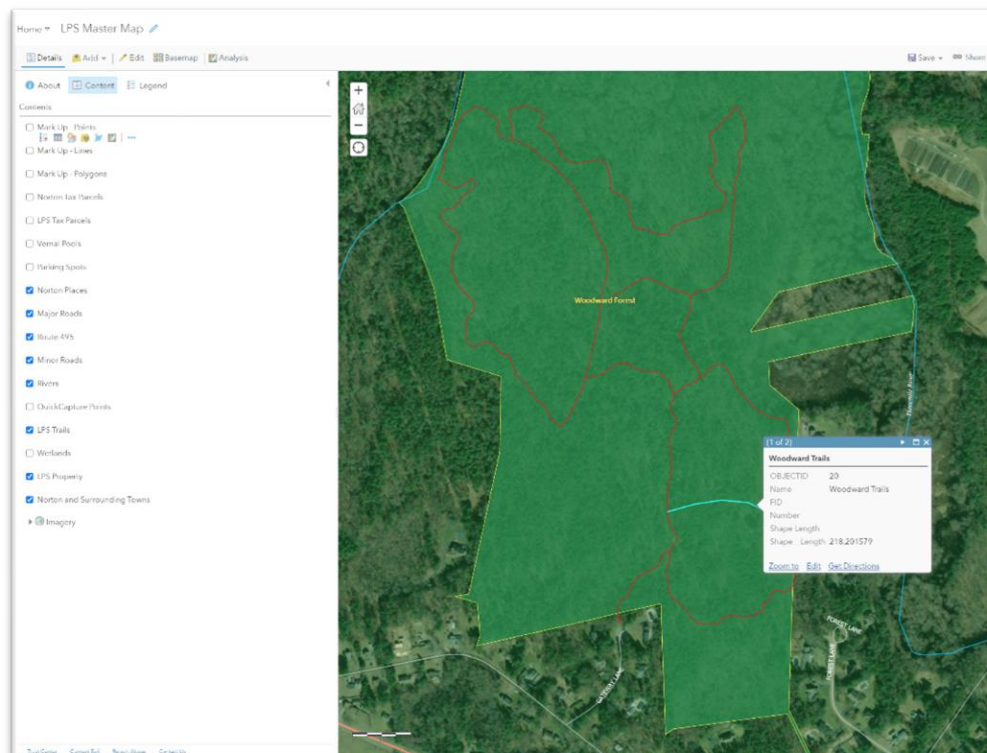
8. You may pause the stream at any time by pressing “PAUSE”. When you continue streaming after a pause, the last collected vertex will connect with the first vertex collected when you resume. It is important to avoid this when mapping noncontinuous trails.

9. When paused, you may also delete the collected line or individual vertices (before sending the update) by pressing the feature you wish to delete and then using the delete buttons at the bottom of the screen. When done streaming, press “PAUSE”, and then press the check mark in the upper left corner.



Note: Updates are automatically sent to the ArcGIS Online Master Map, where they may be edited directly. After updates are sent, the trail may still be edited or deleted through the Collector Classic app. Trails may also be edited using the Land Editor app.

10. When the updates are sent, the recorded line will format the same way the rest of the trails are formatted, and the line may be deleted by selecting it in the Collector viewer and pressing the associated name card at the bottom of the screen, and then the trash can icon. Similar edits can be made through the web map.



Land Department

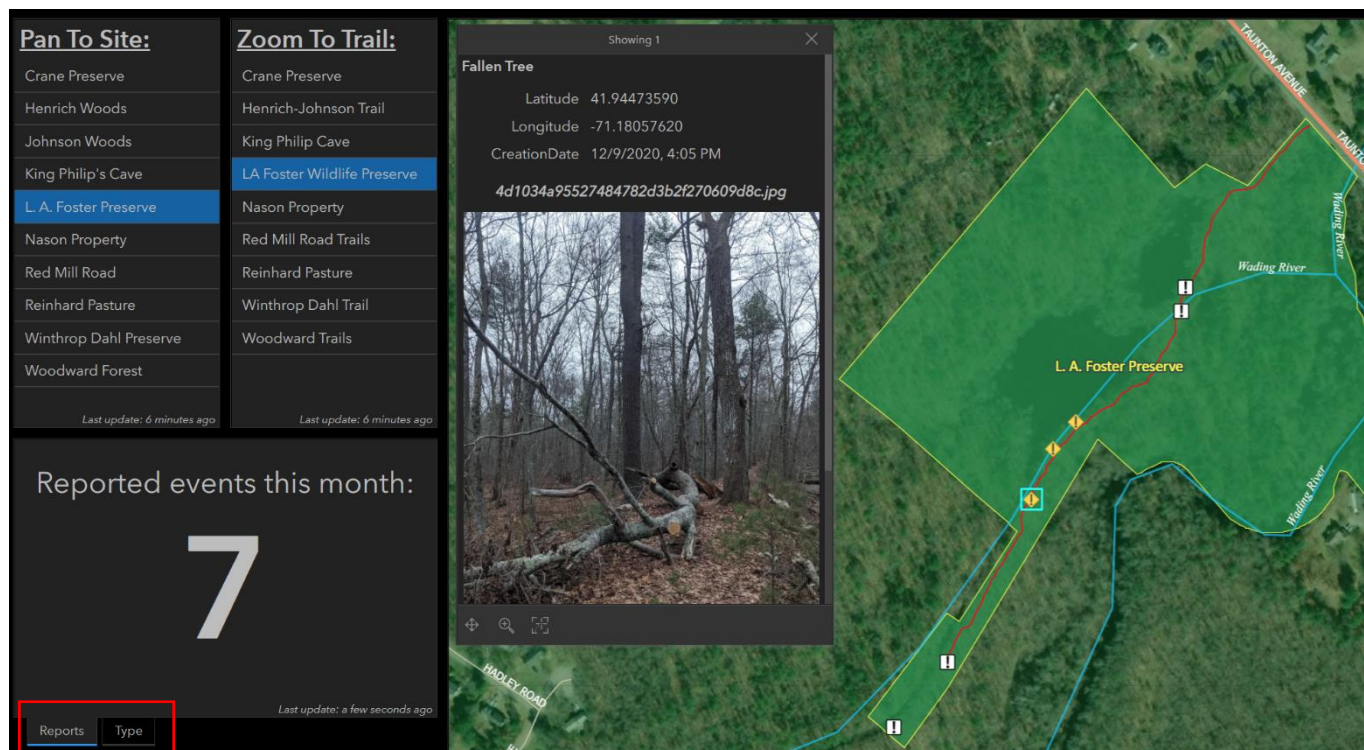
The Land Department of LPS is responsible for managing all aspects of land acquisitions and land management. Some of the primary responsibilities of this department are maintaining property boundary signage along parcel perimeters and ensuring the accurate representation of LPS lands in a digital environment. By providing this department with a land explorer application, an editor application, and a hub application to contain and easily access these apps, Land members will be able to track and manage LPS locations with relative ease.

Dashboards: Land Explorer

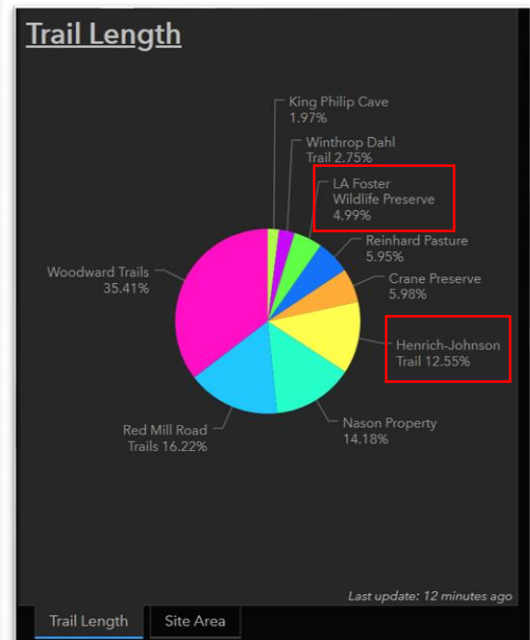
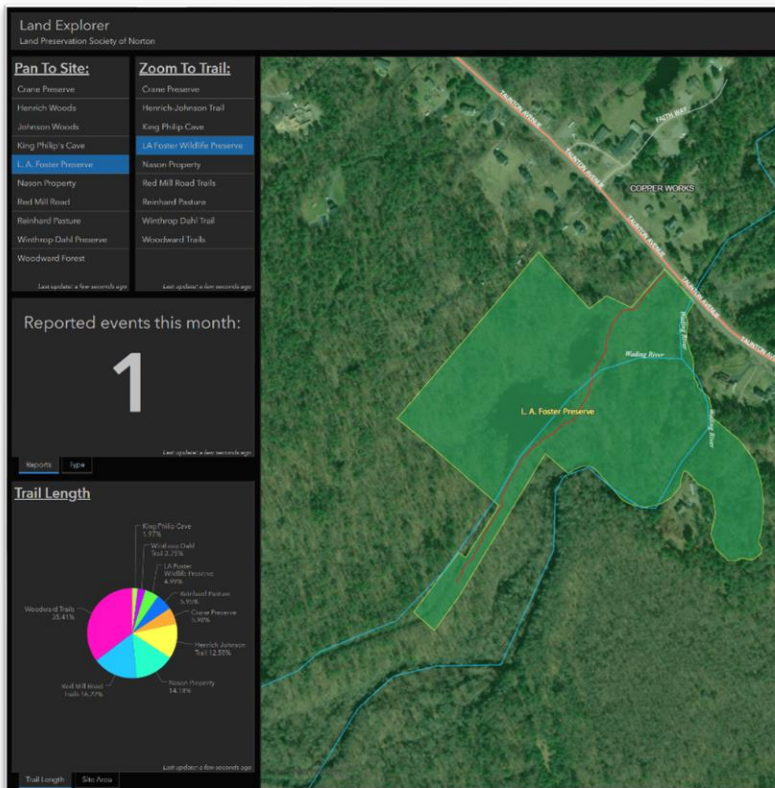
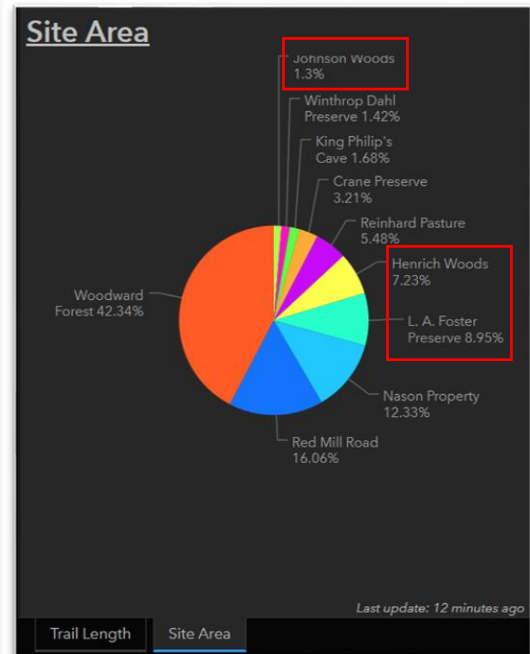
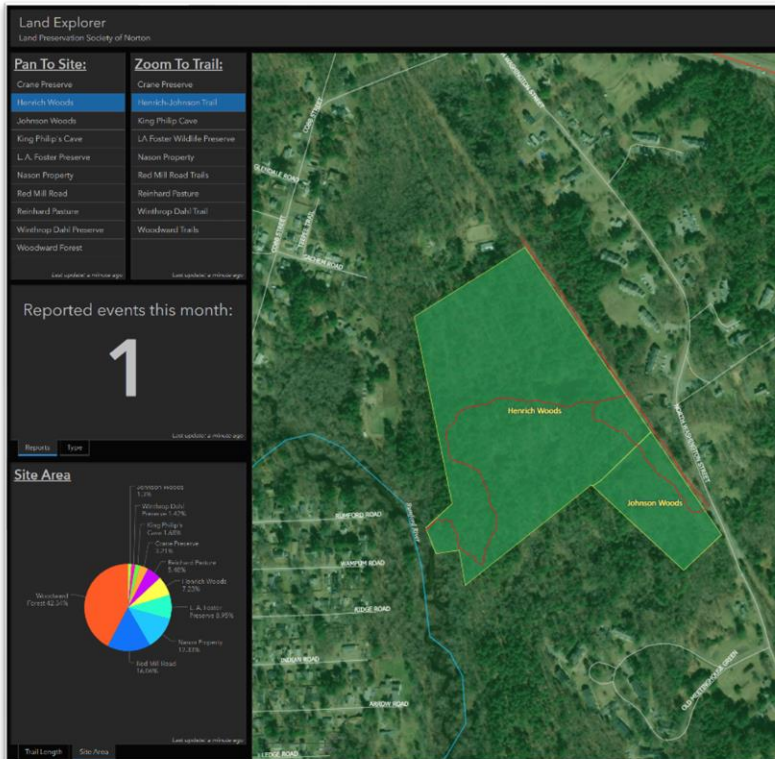
The Land Explorer app is configured as a dashboard layout for use by the Land and Trails departments. It was produced using Web App Builder, and presents layers configured in ArcGIS Pro and ArcGIS Online. The layers include LPS trails and property boundaries for mapped locations, streets and rivers, and point features collected with QuickCapture. The app is optimized to assist the Land department in quantifying features and events reported on LPS land to help inform future acquisitions by allowing members to visualize which locations have the most desirable qualities. This also helps Trails users to update signage on trails where there are hazards or focus attention on producing markers for natural features at sites that have the highest concentrations. The dashboard also displays pie charts showing the proportional lengths of all LPS trails and areas of LPS properties for use in gauging which locations require more attention than others.

Instructions

1. Navigate to the [Land Explorer dashboard](#).
2. Use the “Pan to Site” and “Zoom to Trail” controls in the upper left of the dashboard to navigate around the map. Use the layer selection widget in the upper right corner to turn the “QuickCapture Points” layer and other layers on and off.
3. Refer to the stacked side panel widgets titled “Reports” and “Type” to retrieve statistics on features reported using QuickCapture.



4. Refer to the stacked side panel widgets titled “Trail Length” and “Site Area” to prioritize maintenance tasks and other attention to sites based on their physical characteristics.



By comparing the L.A. Foster Preserve and the Henrich – Johnson Woods, we can observe that there is more property at L.A. Foster, but more trail at Henrich – Johnson. This observation may lead us to conclude that Henrich – Johnson land requires more attention to maintenance, and that L.A. Foster has more potential for trail development.

Web App Builder: Land Editor

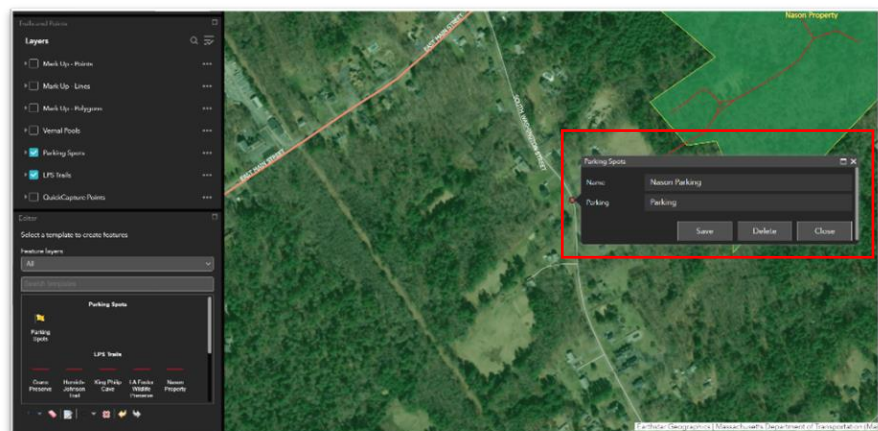
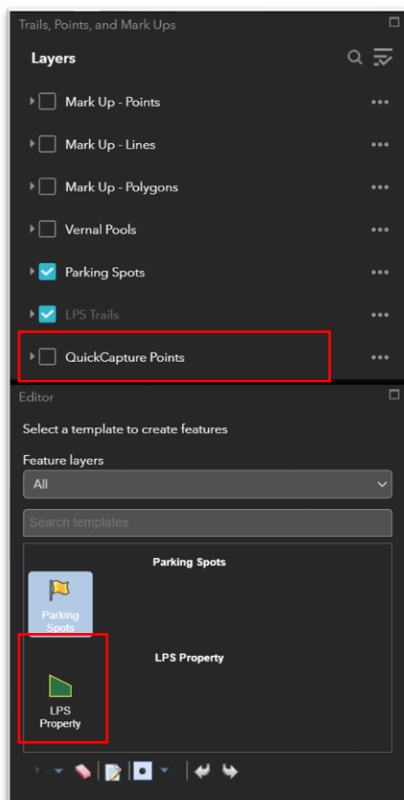
The Land Editor app is designed to enable the Land department to update tax parcel information, to keep track of LPS properties as they are acquired and as they change classification. Side panel widgets allow Land members to markup the map, and to query and export parcels to scout potential land acquisitions. This also helps Trails members to establish trails on newly acquired lands in the future. Trails may also be updated by drawing on the map to connect segments of trails that were missed in the field. Additional work may be done to merge trails in ArcGIS Pro. Points for Parking and Vernal Pools may also be added to the map through the Land Editor.

Instructions

1. Navigate to the [Land Editor web app](#).
2. Use the bookmarks in the upper left corner of the screen to pan between locations.
Note: You may add custom bookmarks to improve navigation efficiency, by pressing “Add”.

To add points, lines, or polygons

3. Select the layer you wish to add to in the *Trails, Points, and Mark Ups* layer selection box in the middle left of the screen. The selected layer becomes visible in the *Editor* widget below it. Click the symbol in the *Editor* to edit it.
4. Zoom to the location where you wish to add the feature, and click.



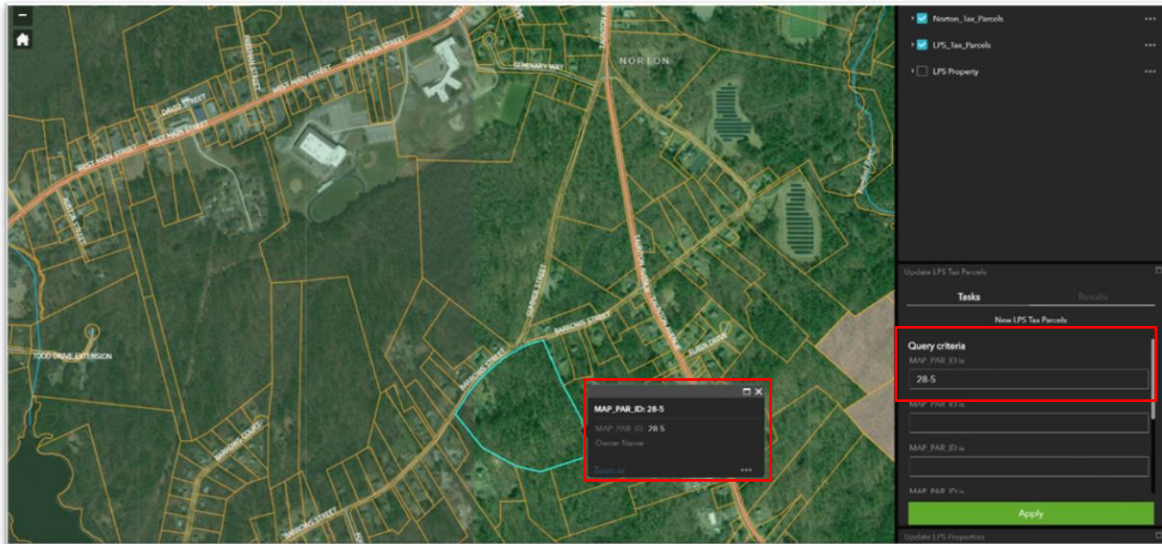
5. Fill in the fields the same way the values for the respective layer are shown in the attribute table and click “Save”.
Note: Refer to the attribute table for the layer you are editing in the Master Web Map for this step. For Mark Ups, no attributes are required.

6. Do not worry about symbol format, it will be consistent with the edited layer when you save.
7. You may also edit the three layers titled *Mark Up* to add points lines or polygons before going into the field, for reference in the Explorer and Collector mobile apps.

To add new parcels to LPS land (the LPS Tax Parcels layer)

8. Select *Norton Tax Parcels* and *LPS Tax Parcels* in the *Parcels* layer selection box in the upper right corner. Deselect *LPS Property*.
9. Zoom to the parcel(s) you wish to add from *Norton Tax Parcels* (up to 5).

- Click the parcel(s) and copy the “MAP_PAR_ID” value(s) to the fields below *Query Criteria* in the *Update LPS Tax Parcels* widget.



- Click “Apply”.
- Notice the result includes all LPS Tax Parcels, and the new parcel(s) you queried. Rapidly select and deselect the new query result added to the layer list above the query widget to see it against the previous extent of LPS Tax Parcels.
- Click the three dots next to the result name in the *Results* tab of the widget, and click “Save to My Content”.



- The new layer is now available in ArcGIS Online and may be used to overwrite the “LPS Tax Parcels” layer.

To update which properties have trails (the *LPS Property* layer)

- Select *Norton Tax Parcels* and *LPS Property* in the *Parcels* layer selection box in the upper right corner. Deselect *LPS Tax Parcels*. Repeat steps 9-13 using the *Update LPS Property* query widget, below the widget used for the previous workflow. Once saved to *My Content* the new layer may be used to overwrite the “LPS Property” layer.

Open Data Hub Site

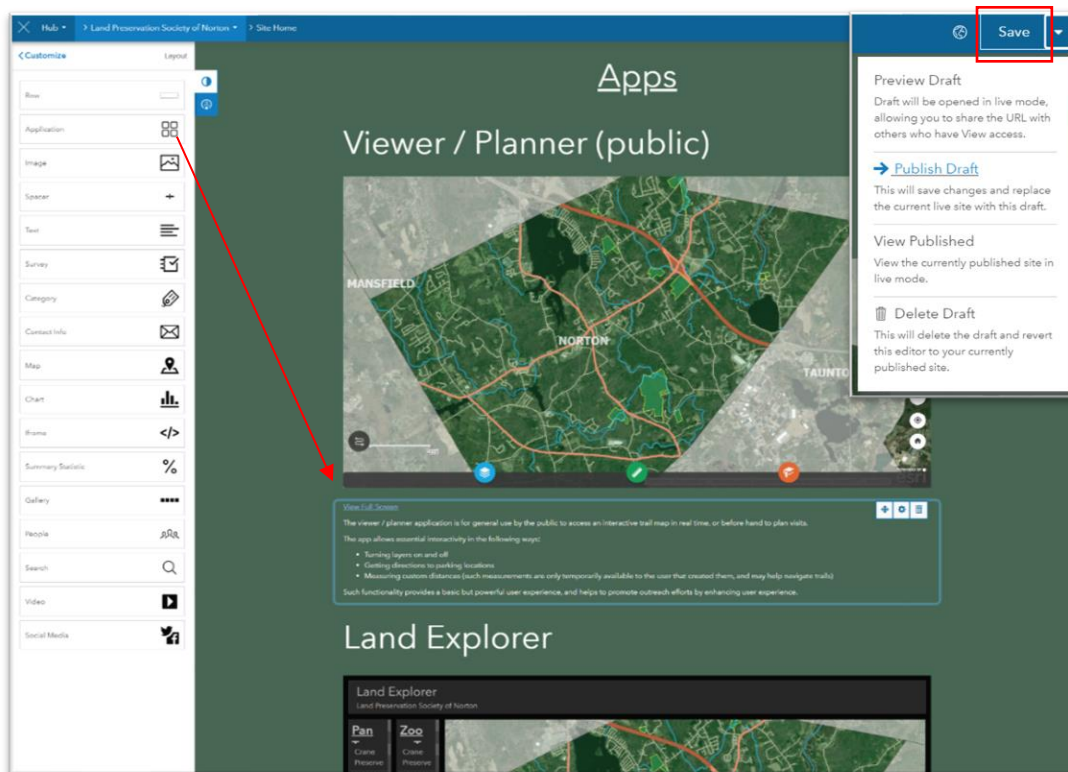
The Hub is intended for broad use in the Land department but may also be referenced by the Trails department to access apps. This app is a centralized container where LPS members can access applications and instructions on how to use them, to perform associated workflows. The Hub may be improved in the future with the addition of other applications as they become available. Currently the Hub contains the Editor application produced in Web App Builder, and the Explorer application produced in Dashboards. Each application includes a functional iframe view, as well as a link to a full screen view and instructions on how to perform associated workflows.

Instructions

1. Navigate to the [Open Data Hub Site](#).
2. Browse through the page to view the Viewer app, the Land Explorer app, and the Land Editor app. Below each app is a link to the full screen version, and instructions on how to use the app. As more apps are produced, the Hub site may be updated to store them.

To Update the Hub Site

3. From ArcGIS Online, select the Hub site in *My Content* and click *Configure App*.
4. Click the pencil icon on the left side of the screen and select *Layout*.



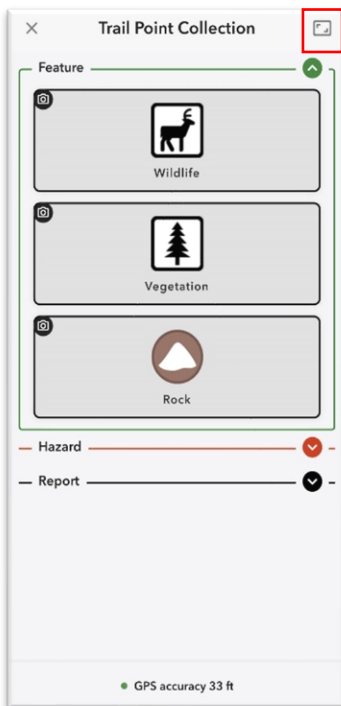
5. Drag a card from the left sidebar to a position in the body of the website. Newly added apps should be situated below the text box of the app above it, or directly above an embedded app.
6. When done editing the site, click the *Save* button in the upper right corner.
7. To publish the site after saving, click the drop-down arrow next to save and click *Publish Draft*. You may preview the draft before publishing by clicking *Preview Draft*.

Outreach Department

The Outreach department of LPS is focused on connecting with the Norton community, and the communities of Wheaton College and other local schools. This area of LPS is the most underdeveloped and may be made stronger and more effective with the implementation of geo apps that help these communities to become involved with Norton land conservation. Nature education is a theme that LPS Outreach frequently uses to spark interest in the organization. By employing a point collector app configured using QuickCapture, involved parties with an ArcGIS login may contribute to a point layer showing various natural features and hazards observed on the trails. A survey produced with Survey123 may also be linked with a QR code at trail heads, fostering public feedback.

QuickCapture

The QuickCapture app will support the Outreach Department by allowing Outreach users to update the public on trail features and hazards. The interface is a simple list of features that may be collected by touching an associated button and taking a picture of the feature to report. This aids in the production of trail reports, helps LPS Outreach to connect with the communities that use the trails, and promotes a safer user experience. It may also be used to involve college students who have an ArcGIS login, helping to foster GIS education and access to local conservation lands.



Instructions

1. Open or install the QuickCapture app and sign in using LPS credentials.
2. Click the “+” icon in the bottom right corner of the screen and scan the associated QR code to add the project in your workspace. Open the project.
3. While visiting a trail site, use the available buttons to report observed features and/or hazards from the given categories when you come across them. You are required to report an image of each thing you observe.
Note: The map button in the upper right corner provides a view of your location and sent points.
4. When you have found a feature to report and taken a picture, press the *Done* button in the bottom right corner.
5. To send collected data, press the map button in the upper right corner of the mobile app and press *Send*. You may also tap a collected point and press *Delete* to remove it. The “QuickCapture Points” layer is updated dynamically to include newly sent data and is viewable in all apps configured with the Master Web Map.



Note: The collected data may be analytically reviewed in the Land Explorer dashboard app, and points may be edited in the Land Editor web app.

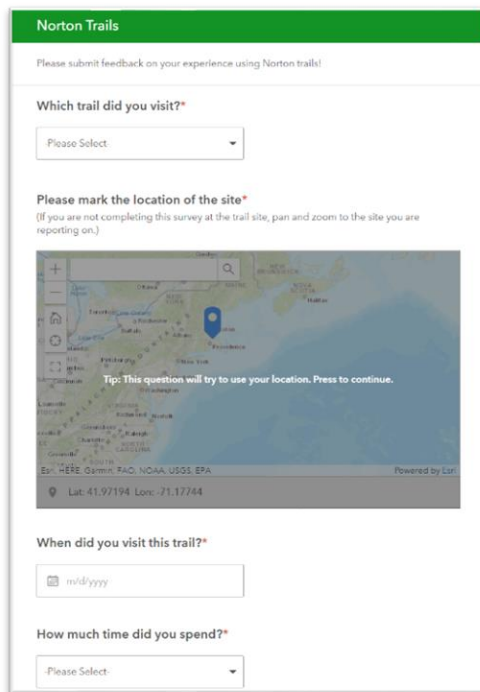
Survey123

The Survey123 app will be one of the most effective tools used to improve the goals of LPS Outreach by allowing for user feedback. The survey is designed to collect feedback both from LPS members and from the public. The survey will collect data on which site was used, quality of experience, and suggestions for improvement. Public responses on trail conditions and hazardous locations may also be submitted. This provides a public means of reporting on trail features and conditions, helping Trails members to target sites for deploying QuickCapture.

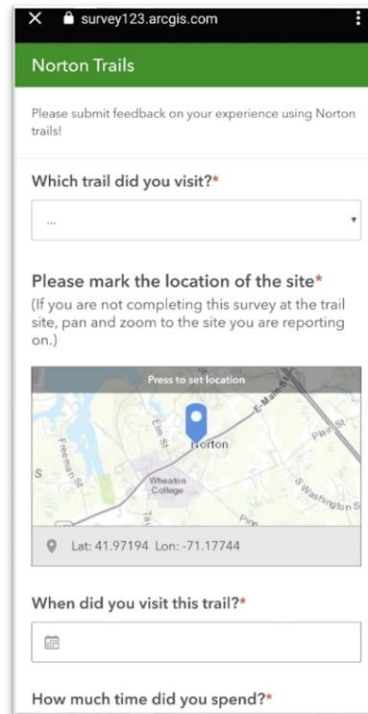
To access and complete the survey

Instructions

1. Navigate to the [online survey](#) or scan the associated QR code to access the survey on mobile.



The screenshot shows the 'Norton Trails' survey form on a desktop browser. The form has a green header with the title 'Norton Trails'. Below the header, there is a prompt: 'Please submit feedback on your experience using Norton trails!'. The first question is 'Which trail did you visit?*' with a dropdown menu labeled 'Please Select.'. The second question is 'Please mark the location of the site*' with a sub-prompt: '(If you are not completing this survey at the trail site, pan and zoom to the site you are reporting on.)'. Below this is a map of Norton, MA, with a blue location pin. A tip says: 'Tip: This question will try to use your location. Press to continue.' Below the map is a text input field for 'When did you visit this trail?*' with a date picker icon and 'm/d/yyyy'. The final question is 'How much time did you spend?*' with a dropdown menu labeled 'Please Select.'.



The screenshot shows the 'Norton Trails' survey form on a mobile device. The form has a green header with the title 'Norton Trails'. Below the header, there is a prompt: 'Please submit feedback on your experience using Norton trails!'. The first question is 'Which trail did you visit?*' with a dropdown menu. The second question is 'Please mark the location of the site*' with a sub-prompt: '(If you are not completing this survey at the trail site, pan and zoom to the site you are reporting on.)'. Below this is a map of Norton, MA, with a blue location pin. Below the map is a text input field for 'When did you visit this trail?*' with a date picker icon. The final question is 'How much time did you spend?*' with a dropdown menu.

2. Fill out the essential questions (marked with a red asterisk) and optionally fill in the non-essential questions to provide more detailed feedback on the experience.
3. Follow the link at the bottom of the survey to view a public web app of LPS trails.

To view analytics on responses to survey questions

4. Navigate to the [Survey123 website](#) and sign in using LPS credentials.
5. Click the “Norton Trails” survey under *My Surveys* and then click the *Analyze* tab.



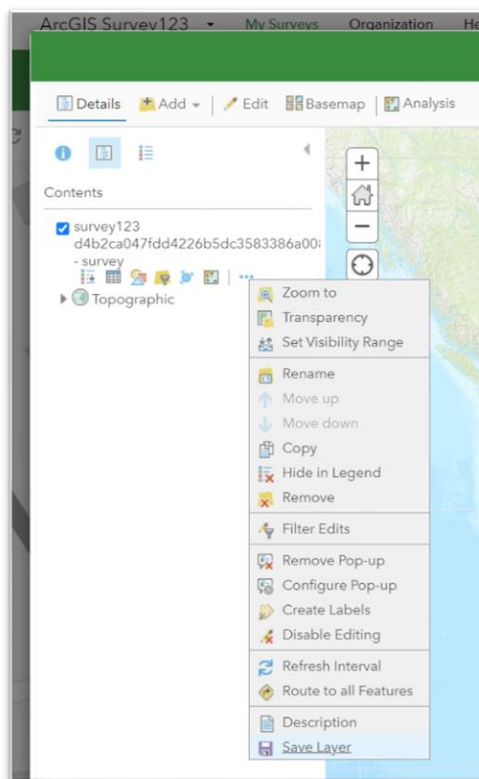
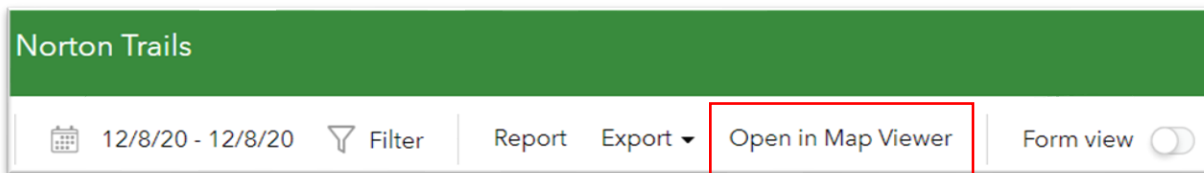
6. Browse through the page to view data visualizations of the responses to each question.

To work with data collected through the survey

7. Navigate to the [Survey123 website](#) and sign in using LPS credentials.
8. Click the “Norton Trails” survey and then click the *Data* tab.



9. To view data on a web map, click *Open in Map Viewer*.
10. From here the survey may be saved to ArcGIS Online as a layer by clicking the three dots next to the layer and clicking *Save Layer*. Provide a title and tags for the layer and click *Save*.



11. You may now configure the survey data in ArcGIS Online.
12. Data may also be exported in various formats for offline use by clicking the *Export* tab, next to *Open in Map Viewer*.

Synthesis and Conclusion

While the apps are binned into groups based on which department they benefit the most, there is overlap and association among them. The Explorer mobile app, while mostly applicable to the goals of the Trails department, may also be used by the Land department to assist in the deployment of property boundary markers. The QuickCapture and Survey123 apps are designed to assist LPS Outreach but may also be used by the Trails department to collect features on trails, and to submit feedback on trail use experience. The Open Data Hub site application is a conveniently accessible storage space for apps used mostly by the Land department, but the Trails department may access this site to use the apps for their purposes of updating line and point information as well. Finally, the Land Editor app may be used by both the Land and Trails departments to update LPS tax parcels and to add point and line data of concern to each department.

With the adoption of this geospatial toolkit, the technology will help to make up for the small population of the organization. This suite of geo applications and workflows will provide them with mapping tools and methods of collecting data on their land that did not exist before. The information derived from these apps may be used to update the LPS website and to create more accurate and decorative trail-head signage. It may also be strategized by students and interns involved with Norton conservation land in the future, and applied in different ways to accomplish related academic and extracurricular goals. It is of great importance to the LPS that the public stay informed and have a safe and accessible user experience while using LPS trails. These methods and technologies provide a vehicle by which trail improvement, land management, and outreach efforts may be safely and more easily approached, and they advance the mission of the LPS.